

Edge AI Deployment Guide

Deploying AI models on edge devices requires careful optimization of model size, inference latency, and memory footprint. This guide covers deployment strategies for NVIDIA Jetson, Apple Silicon, and Qualcomm Snapdragon platforms.

Key considerations:

- Quantization: 4-bit reduces model size 4x with minimal quality loss
- Batching: Process multiple inputs to amortize overhead
- Model pruning: Remove redundant parameters for faster inference
- Hardware acceleration: Metal (Apple), CUDA (NVIDIA), Hexagon (Qualcomm)

Deployment targets:

- NVIDIA Jetson Orin: 275 TOPS, ideal for real-time vision + language
- Apple M4: 38 TOPS Neural Engine, excellent for on-device RAG
- Raspberry Pi 5: CPU-only, suitable for text embedding at small scale